

AgentPProf: Operation-Stack Profiling for AI Agent Traces

Abstract

AI-agent executions are no longer well described by one prompt, one session, or one trace tree. A single run can contain prompts, tool calls, GUI actions, browser actions, process events, plans, subagents, safety labels, failure labels, and human subtask boundaries. Existing observability systems usually make one boundary primary, such as a session, span, or prompt tree. Classic profilers make another boundary primary, such as a function call stack. This paper argues that agent profiling needs a smaller abstraction: an operation is a weighted record with fields, and an operation stack is a recursive projection over those fields chosen at query time. Aggregation is not a separate object or a fixed trace tree; it is the result of choosing a stack shape over the same operations. This is not merely a claim about query-time aggregation, which classic profilers and trace processors also support in different forms. The scoped novelty is the combination of an agent-operation record model, recursive multi-field operation-stack projections, and a hidden-label cross-dataset localization benchmark.

We implement this model in the Rust AGENTPPROF profiler and evaluate it on public labeled agent trajectories. Across 15 public sources, the profiler maps 47,590 samples into the same operation layer. On four oracle-rich families, AgentRewardBench, SATraj-OS, AgentNet, and OSWorld-Human, we build six localization tasks over 34,539 operations and 3,699 positive labels. The main R320 experiment treats profile groups as ranked localization outputs and scores 144 view/ranker policies with hidden labels. Query-aware operation-stack profiling inspects a median 9.37% of operations in the top five groups, compared with 100% for flat summaries. Against fixed-session drilldown, operation stacks improve top-five recall on 5 of 6 tasks and reduce median group fragmentation from 285.0 to 157.5 groups. Query-aware ranking improves average precision over width-only operation-stack ranking on all six tasks. The Rust profile-spec implementation of operation-level rank features improves semantic-stack AP over width on 5 of 6 tasks. A reproducibility probe reruns 76 tracked profile specs and an opt-in deterministic output mode makes all 76 raw profile artifacts byte-stable across repeated runs. A held-out transfer probe selects rank policies without using target-task labels; leave-task and leave-dataset selection both improve semantic/coarse AP over width on 4 of 6 tasks and semantic first-positive work on all six tasks. A paired-bootstrap uncertainty audit over the same six task families finds stable directions for the main AP, work, recall, and fragmentation comparisons while preserving flat-recall and fixed-session-work counterpoints. A label-permutation negative-control audit keeps visible rankings fixed and shows that operation-stack query-aware AP exceeds the prevalence/group-size null on all six tasks, while top-five precision and first-positive work remain limited. A view/depth task-fit audit shows that best visible AP is split across operation-stack, fixed-session, and dataset-native views, and that best top-five F1 spans five view families; no single hierarchy or source-task selector is enough. An inspection-efficiency audit reruns the R320 scorer over tracked operation inputs and shows that, within a 30% inspected-work budget, query-aware operation stacks reach median recall

0.3900, versus 0.0000 for flat summaries, 0.3559 for fixed-session drilldown, 0.3377 for dataset-native views, and 0.3325 for raw action stacks. A fragmentation audit separates group fragmentation from operation work: operation-stack query-aware reaches 50% positives with fewer ranked groups than fixed-session query-aware on 5 of 6 tasks and inspects fewer groups at the same 30% work budget on 5 of 6 tasks, while lowering work-to-first-positive on only 2 of 6 tasks. An actionability synthesis then merges the accuracy, ablation, robustness, transfer, view-depth, and fragmentation audits into six task cards: all six contain concrete optimization actions and AP gains from query-aware ranking, while mapping, rank features, stack depth, transfer policy, and fixed-session drilldown remain task-specific tradeoffs. A multi-objective selection audit then scores 15 visible policies over six diagnostic objectives: operation-stack query-aware remains Pareto-frontier on all six tasks and keeps lower top-five work than flat and higher top-five recall than fixed-session on the main comparisons, while every task still has multiple objective-specific best policies. A fixed-recall target audit further shows that operation-stack query-aware reaches a 25% positive-recall target on all six tasks with median work 0.2000 and median 16.0 groups; flat reaches the same target only at median work 1.0000, and fixed-session reaches it with median 50.0 groups. A sequence-scope adequacy audit then scores whether ranked groups hit trajectories that actually contain positives. At a 30% operation budget, operation-stack query-aware reaches median 0.4669 positive-session recall versus 0.3230 for fixed-session while touching median 0.3467 sessions; raw action reaches 0.5147 positive-session recall but touches 0.9103 sessions. An oracle-depth adequacy audit then scores the same visible-ranked profiles at the depth provided by each dataset: session, operation or step, positive-run proxy, and task-specific units such as OSWorld-Human human groups. Across 24 task-depth rows, operation-stack query-aware uses lower top-five oracle-unit work than flat on 24 of 24 rows, improves 30%-budget positive-unit recall over fixed-session on 20 of 24 rows, improves unit F1 on 18 of 24 rows, and reaches 50% positive units with fewer groups on 22 of 24 rows. The audit also records a depth-gap counterpoint: it does not reduce positive sessions without a hit unit relative to fixed-session. A cross-task and cross-dataset policy-transfer audit then selects visible policies from non-target tasks or non-target datasets before held-out scoring. Across 96 decisions, selection stays within tolerance on 62 decisions and beats width, fixed-session, and flat baselines on 72, 69, and 41 decisions. Since operation-stack views are selected on only 16 decisions, the result supports objective-specific policy transfer rather than a universal default view. Finally, a mechanism-attribution audit classifies the actionability evidence: 36 of 36 objective-task recommendations have concrete optimization actions, 27 of 36 best visible policies are not the default operation-stack view, and 34 of 96 transfer decisions outside tolerance are assigned view/ranker or baseline-counterpoint explanations. A profile-spec composition audit then checks the concrete operation-stack composition path over the same real-trace Rust outputs: 12/12 variants compose operation files, predicates, operation-level rank rules, rule-score ranking,

and explicit stack depth without prompt/session frames; visible operation-feature ranking improves AP over width on 9/12 variants and coarse depth reduces groups on 6/6 tasks with median reduction 0.8267. Finally, a metric-consistency audit checks that this is a multi-metric tradeoff rather than an AP-only result: 30/50 baseline-metric comparisons support the operation-stack query-aware policy, 16/50 are explicit counterpoints, and nDCG and coarse top-k recall remain secondary metrics. R345 then turns actionability into a diagnostic-lens portfolio: 6 diagnostic lenses cover 36 objective rows, operation-stack-family views win 11/36 objectives, non-operation-stack counterpoints win 25/36, and 6/6 tasks need at least three best views. This supports lens-specific optimization, not a single flamegraph or universal selector. R346 then turns the top-ranked operation-stack groups into a diagnostic casebook: 30 case groups, positives in top-5 groups on 6/6 tasks, positives in top-1 groups on 5/6 tasks, median top-5 lift 1.6508, and median top-5 work 0.0937. This gives case-level evidence for the actionability claim without claiming human utility. R347 then contrasts the same case-level evidence against 5 visible views: flat, fixed-session, dataset-native, raw-action, and operation-stack. Operation stacks win against flat top-5 work on 6/6 tasks, fixed-session top-5 recall on 5/6 tasks, and fixed-session group count on 4/6 tasks, while fixed-session first-positive work and flat full-work recall remain counterpoints. R348 then converts actionability into objective-level counterfactuals over already-scored visible policies: 36/36 best rows are visible non-oracle, 27/36 objective rows require non-default actions, 25/36 require view changes, and 2/36 require operation-stack ranker tuning. The median gain over the default operation-stack policy is 0.1447, while 6/6 tasks expose at least three action classes plus case counterpoints. R349 then maps held-out R340 policy transfer decisions to R348 action classes. Across 60 aligned decisions, held-out selection is within tolerance on 35/60 and beats the default on 30/60, but exact action-class transfer is only 7/60 and only 2/42 for non-default target actions. This supports protocol-sensitivity and actionability tradeoffs, not an automatic action selector. R350 combines the casebook, baseline contrast, action-counterfactual, and held-out guardrail artifacts into bounded evidence packets. Top-five operation-stack packets contain positives on 6 of 6 tasks, top-one packets on 5 of 6, and 4 of 6 tasks meet the strict 30% operation-work budget; the same packets preserve 27 of 36 non-default objective rows and the 35 of 60 within-tolerance / 7 of 60 exact-action transfer guardrail. R354 turns one such actionability path into executable before/after profile specs, improving AP, top-five lift, and first-positive work on 5 of 6 tasks; R355 adds the oracle-depth scorer described above and preserves the fixed-session depth-gap counterpoint. A paper claim-integrity audit then checks the paper-facing R320–R350 numbers, source policy, guardrails, and the two-abstraction boundary across the Chinese and English drafts. The result-invariant, source-policy, and guardrail checks pass. A final evaluation-rubric audit maps the same evidence to an OSDI-style profiling-paper standard. It reads tracked R320–R351 artifacts and the current paper text, downloads no data, reruns no profiler, and passes 26/26 required checks across ten rubric areas. The audit classifies the current claim as a scoped profiler-benchmark gate, not evidence for human/agent analyst utility, automatic boundary discovery, or full trace-ecosystem compatibility. The result supports a

profiling claim rather than a user-study claim: operation/operation-stack profiling can localize and rank task-relevant failures, quality problems, and semantic boundaries against dataset-provided labels in this benchmark, while exposing actionable tradeoffs among stack fields, mappings, rankers, and drilldown depth.

1 Introduction

AI-agent traces mix many objects that traditional profiling and tracing systems keep separate. A coding agent may ask a model for a plan, call a shell tool, launch a subagent, inspect a file, and retry a failing command. A web or desktop agent may click, type, observe a page, repeat an action, and receive a benchmark label saying that the action was unsafe, redundant, incorrect, or the start of a human subtask. If the profiler fixes the hierarchy at prompt, session, or span boundaries, these questions become hard to ask: which phases concentrate unsafe operations, which action families cause redundant steps, and which groups of operations cross sessions but share the same failure mode?

AGENTPPROF takes the opposite position. The profiler stores a sequence as operations, where prompts, sessions, tool calls, processes, syscalls, GUI actions, and plan steps are all operation shapes or operation fields. Users then choose an operation stack to fold the same operations at different depths. One query may group by task, phase, action; another may group by environment, safety. Mapping and tagging are first-class field-derivation steps, but they do not add a third profiler object. They only write fields before the stack query runs. Thus the novelty is the abstraction that connects stack shape to aggregation: changing the stack changes what is comparable, localizable, and actionable without changing the underlying trace object model.

This paper makes a profiling claim, not a human-productivity claim. A top-tier profiling paper must show that the profiler attributes, localizes, and ranks problems against explicit workload labels, and that the output leads to optimization insight with better label-scored ranking or inspection cost than baselines. For agents, the claim is:

Operation/operation-stack profiling provides label-scored localization and ranking for task-relevant failures, quality problems, and semantic boundaries in real labeled agent traces, while requiring less inspection work than flat summaries and less fragmentation than the fixed-session drilldown used here as a span-tree proxy.

We evaluate this claim with public labeled traces rather than a human study. A human or agent analyst study would be useful for productivity claims, but it is not required to test label-scored profiler fidelity. Our evaluation therefore uses hidden labels as the oracle for ranked profile groups and is organized around four core experiments. E1 tests operation coverage, recursive stack folding, field derivation, and human-boundary alignment under the same two-object model. E2 is the main hidden-label localization and ranking benchmark. E3 isolates mechanisms and actionability through ranker, stack, mapping, transfer, case, and profile-spec patch ablations. E4 checks reproducibility, offline cost, and claim-integrity guardrails. Individual R-runs are provenance for these experiments, not the paper-facing structure.

2 Design

2.1 Operations

An operation is the only sample-level object in AGENTPPROF. It is a weighted record with string fields. The same representation can encode a prompt, a tool call, an API call, a browser action, a PyAuto-GUI action, a process event, or a derived label such as `looping=yes` or `step_correct=incorrect`. External datasets enter the system as operation JSONL. Local Codex or Claude sessions can also be exported as an agent-session trace, imported back, or converted to operation JSONL before profiling.

2.2 Operation Stacks

An operation stack is an ordered list of fields. Folding operations by that list yields the usual folded-stack representation, but the stack is chosen at query time. The same operation file can be folded as a flat task summary, a fixed-session tree, a dataset-native hierarchy, a raw action stack, a semantic operation stack, or an oracle label-drilldown. Fixed sessions and spans are therefore baselines and exchange containers, not core profiler abstractions.

2.3 Mapping and Tagging

Mapping and tagging derive new operation fields before folding. For example, desktop actions such as `type` and `key` can map to `phase=input`, while `tool/API` rows can map to a `tool phase` before generic action rules run. The Rust CLI exposes inline rules, rule files, profile specs, operation predicates, and stack overrides. `-where` predicates run after mapping and before stack construction, implementing the query predicate without adding a profiler object. Profile specs record operation files, mappings, predicates, views, stacks, and outputs for reproducibility, while command-line flags can still override the stack query.

3 Evaluation Setup

Table 1 summarizes the public trace families. We do not create new test sets, sync complete image corpora, or depend on gated data. The 15 sources establish generality, while the main hidden-label ranking results use the four oracle-rich families with failure, safety, quality, and human-boundary labels.

For R320, we build six tasks: AgentRewardBench looping, AgentRewardBench side-effect, SATraj unsafe operation, AgentNet incorrect step, AgentNet redundant step, and OSWorld-Human group start. Each task has an oracle-positive field, but non-oracle rankers never read that field. We compare six views: flat, fixed-session, dataset-native hierarchy, raw action stack, semantic operation stack, and label-drilldown. We compare four rankers: width, visible-risk, query-aware, and oracle upper bound. Label-drilldown and oracle upper bound are marked as hidden upper bounds.

The metrics match a profiler paper’s main claim. Fidelity uses `precision@k`, `recall@operation-budget`, `F1@k`, average precision as a block-level AUPRC-style score, nDCG over positive operation counts, and work-to-first-positive. Fragmentation uses group count, positive group count, and work or groups to reach 50% positive recall. Actionability uses per-task comparisons among stack fields, mapping choices, ranker choices, and oracle headroom. Reproducibility uses repeated Rust `agentpprof -profile-spec`

executions over tracked specs and tracked operation JSONL inputs, checking semantic output hashes, sample counts, unique-stack counts, runtime, and output size. Build time is excluded from the profiler runtime measurement.

Table 2 gives the paper-facing evaluation structure. R-numbered runs are provenance, ablations, counterpoints, or audit gates rather than the paper’s main organization. R360 regenerates this four-row result table from tracked artifacts and emits JSON, CSV, Markdown, HTML, and a LaTeX table fragment. It is a table-provenance gate, not a new empirical result or profiler rerun.

4 Results

4.1 E1: Coverage and Recursive Folding

Evidence contract: claim, the operation/operation stack model covers heterogeneous traces without prompt/session/tool-bound units; oracle, public labels and native fields, especially OSWorld-Human boundaries and AgentNet quality; baseline, dataset-native/no-map/fixed-session stacks plus replay and trace round trips; metric, coverage, stack range, compression, boundary F1/V-measure, and equality; counterpoint, configurable operation stack folding, not complete compatibility or latent-intent recovery. Prompt, session, tool, process, and syscall records are operation fields, not new profiler objects.

The 14 core public sources produce 42,590 operations. Adding the supplemental ScaleCUA stream brings the smoke set to 47,590 operations. These operations cover web, mobile, desktop, software, API, tool-agent-user dialogue, expert failure labels, safety labels, human grouped-action labels, and desktop step-quality labels. No source required a prompt-specific, GUI-specific, or safety-specific profiler object.

Recursive stack-depth sweep. On the same 13,265 operations, a depth sweep folds to 9 dataset stacks, 57 phase stacks, 226 tool/semantic stacks, 455 action stacks, and 3,757 fixed-session stacks. A fixed session boundary therefore increases stack count by 417.4× relative to dataset depth. On AgentNet, a tracked profile spec folds 16,741 operations into 608 diagnostic stacks, while a CLI stack override changes the output to 83 stacks without changing the operation input. The result supports query-time recursive stack choice rather than a fixed prompt/session hierarchy. R321 adds an implementation probe for query predicates: profile specs over the tracked R300 operation file select 729/729, 714/714, and 4,285/4,285 expected operations with mapping-derived `where_rules` before folding. R322 adds the corresponding Rust JSON ranking surface: profile specs with visible `rank_rules` emit ranked operation-stack groups over the same six tasks, improving AP over width on 4 of 6 tasks while preserving counterexamples for richer group-level rankers. R323 adds a `rank_mode` probe: score-first ranking improves AP over width-boost on 4 of 6 tasks and reduces SATraj first-positive work from 0.6376 to 0.0842, while side-effect and OSWorld-Human remain counterexamples. R324 adds operation-level rank features: `rank_op_rules` match visible mapped `field=value` operation tokens and aggregate matched weight inside each folded group. On semantic stacks, AP improves over width on 5 of 6 tasks and first-positive work improves on 5 of 6 tasks; on coarser stacks, AP improves on 4 of 6 tasks while group counts fall. R325 replays the

Table 1: Public labeled trace families used by the evaluation. The main R320 hidden-label profiler benchmark uses the four oracle-rich rows in the bottom half.

Family	Native labels	Access path	Evaluation role
WebLINX, Mind2Web, AgentTrek, WebShop	Web actions, demos, tasks, rewards, action history	Hugging Face viewer or public shards	Web navigation coverage
API-Bank, ToolBench, taubench	API calls, tool messages, solution paths, outcomes	Hugging Face rows or repo JSONL	Tool/API and dialogue coverage
SWE-agent	Issue trajectory, command actions, target outcome	Hugging Face viewer	Software-agent coverage
AndroidControl, Odyssey, ScaleCUA	Mobile or GUI actions, instructions, history context	Public/HF samples	Mobile and GUI coverage
AgentRewardBench	Expert success, side-effect, looping, optimality	HF annotations plus cleaned BrowserGym steps	Failure and looping tasks
SATraj-OS	Desktop actions, safety, reward, attack type	HF safety config	Safety task
OSWorld-Human	Human single-action and grouped-action traces	Public GitHub JSON	Human-boundary task
AgentNet	Human desktop PyAutoGUI, step correctness, redundancy	HF repo JSONL streaming	Step-quality tasks

Table 2: Paper-facing core experiments and generated headline evidence. R360 regenerates the table from tracked artifacts; individual R-runs remain provenance and counterpoint records.

Core experiment	Workload	Generated headline evidence	Scoped conclusion
E1: coverage and recursive folding	15 public labeled trace families / 47,590 operations, plus local-session and standard-trace exchange fixtures	R286 folds the same 13,265 operations across 8 depths, from 9 to 3,757 stacks; R285 has 6/9 positive leave-dataset-out stack reductions; R342 has 12/12 prompt/session-free profile specs; R353 round-trips 512 real operations with 512 samples and 11 stacks preserved.	Two abstractions cover heterogeneous traces; trace containers and mappings are not profiler objects.
E2: hidden-label localization and ranking	Six oracle-backed tasks over AgentRewardBench, SATraj-OS, AgentNet, and OSWorld-Human	R320 covers 6 tasks, 4 datasets, 34,539 operations, 3,699 positives, and 144 policies. Operation-stack query-aware uses 0.0937 top-five work versus flat 1.0, beats fixed-session top-five recall on 5/6 tasks, and uses 157.5 median groups versus fixed-session 285.	Supported as a hidden-label profiler benchmark with baseline tradeoffs, not human utility.
E3: mechanism and actionability	The same six tasks plus held-out OSWorld-Human boundary-backend operations	R354 accepts 5/6 executable profile-spec patches, with median AP delta 0.0376 and top-five lift delta 0.5750. R358 improves OSWorld-Human boundary AP to 0.2583 versus 0.2402 and reduces groups to 74 versus 108, while preserving top-five-work and first-positive-work counterpoints.	The profiler exposes stack, field, ranker, and profile-spec knobs; it is not an automatic selector or boundary detector.
E4: reproducibility, cost, and claim integrity	Tracked profile specs, paper/docs hashes, and claim-scope gates	R328 gives 76/76 semantic and 76/76 raw-byte deterministic specs over 152 invocations, with median runtime 1.601s and p95 2.767s. R338 has 350/350 number checks; R352 reaches level_4_scoped_profile_benchmark; R357 has 4/4 reviewer ACCEPT; R359 has 13/13 core-table checks.	Supported as reproducible offline profiling-paper evidence, not live overhead or complete trace-ecosystem compatibility.

same scrubbed profiler input and stack projections with newly generated profile specs that remove one visible feature at a time. It finds 7 critical feature instances, 3 misleading feature instances, and a stack-depth tradeoff: coarse stacks are AP-preferred on 2 of 6 tasks but reduce group counts on all six tasks. R326 tests whether these rank policies are brittle hand-tuned weights. A global equal-weight visible feature bank improves AP over width on 4 of 6 semantic tasks and 5 of 6 coarse tasks. Equal-weight task policies stay within 0.02 AP of their weighted counterparts on 8 of 12 task/depth variants. R325-guided repairs improve AP on 2 of 3 misleading-feature cases and first-positive work on 2 of 3 cases; 1 of 3 improves both metrics. This is post-hoc actionability evidence, not a label-free deployment ranker. R329 adds a target-held-out transfer probe. Candidate policies include the global equal feature bank and six source-task equal-weight policies. Leave-task and leave-dataset selection use only non-target task scores, while target labels are reserved for final scoring. Both protocols improve semantic/coarse AP over width on 4 of 6 tasks and semantic first-positive work on 6 of 6 tasks, but side-effect, SATraj, and boundary counterexamples remain; the result isolates policy transfer from target-label tuning without claiming a universal ranker.

Field derivation and boundary probes. Held-out mapping rules improve compression from 14.049 to 19.091 on 3,990 operations. Leave-dataset-out mapping reduces unique stacks on 6 of 9 held-out datasets with no negative stack-reduction regressions after API/tool phase precedence is fixed. A supervised OSWorld-Human boundary backend reaches F1 0.7735 on held-out adjacent pairs and writes predicted boundary fields that the Rust profiler folds as ordinary operation fields. Boundary backends therefore extend field derivation, not the profiler object model.

Human-boundary oracle. OSWorld-Human gives the strongest direct boundary oracle. On 320 exact-aligned tasks with 4,011 operations and 2,075 human groups, grouped-depth stacks reduce 482 action stacks to 106 grouped stacks. A conservative group_pattern projection has human-group boundary F1 0.627 with precision 1.0 and recall 0.456. This is not complete intent discovery, but it shows that the same single-action sequence can be folded at action depth or human-group depth and scored against a human boundary oracle.

4.2 E2: Hidden-Label Localization and Ranking

Evidence contract: claim, operation-stack profiling localizes and ranks real labeled failures, quality problems, safety issues, and semantic boundaries; oracle, 3,699 positives over 34,539 operations; baseline, flat, fixed-session/span-tree proxy, dataset-native, raw-action, operation-stack, label-drilldown, and oracle upper bound; metric, precision@k, recall, F1, AUPRC-style AP, nDCG, budget recall/F1, work-to-first-positive, and groups; counterpoint, flat and dataset-native views can win broad-recall or nDCG goals, so the claim is Pareto tradeoff rather than metric dominance.

Table 3 gives the main R320 accuracy result. Flat summaries recover all positives, but only by inspecting the whole task, so their median work@5 is 1.0 and their work-to-first-positive is 1.0. Fixed-session drilldown finds an early positive with median work-to-first-positive 0.0044 and median work@5 0.0163, but its median recall@5 is only 0.0233 and it fragments the workload into 285.0 groups. Query-aware operation stacks inspect median 0.0937 work in the top five groups, reach median recall@budget30 of 0.3900, and reduce the median group count to 157.5.

The strongest comparison is not universal dominance. Operation stacks improve top-five recall over fixed-session query-aware drilldown on 5 of 6 tasks and reduce group fragmentation from median 285.0 to 157.5 groups, but fixed-session often finds the first positive with less work. Operation stacks also save work against flat summaries on all tasks, but flat remains a complete full-recall counterpoint. Dataset-native hierarchy sometimes has higher recall, but at high inspection work. These mixed results are the desired profiling-paper evidence: operation stacks occupy a useful Pareto region between flat summaries and fixed-session drilldown.

R330 audits uncertainty over these R320 comparisons without rerunning the profiler or creating data. It bootstraps paired task-family deltas for 10,000 resamples. Against flat width, operation-stack query-aware has stable positive AP delta (mean 0.1219, 95% CI [0.0406, 0.2175]), stable lower top-five work (mean -0.8168, CI [-0.9653, -0.6568]), stable higher 30% budget recall (mean 0.4510, CI [0.3450, 0.6143]), and stable lower work-to-first-positive (mean -0.9303, CI [-0.9855, -0.8650]). Against fixed-session query-aware, operation stacks have stable higher top-five recall (mean 0.1801, CI [0.0542, 0.3127]), higher top-five F1 (mean 0.1702, CI [0.0549, 0.2870]), and fewer groups (mean -178.0, CI [-340.0, -39.0]). Against width-only operation stacks, query-aware ranking has stable higher AP (mean 0.0932, CI [0.0110, 0.2184]) and lower top-five work (mean -0.3101, CI [-0.4209, -0.1997]). The same audit marks flat top-five recall, fixed-session first-positive work, and raw-action mapping comparisons as mixed or counterpoints; it is a task-level uncertainty audit, not a per-operation independence or human-utility claim.

R331 adds a prevalence and group-size negative control. It keeps each visible policy's groups and ranking order fixed, then randomly reallocates each task's hidden positive operations across same-size group positions for 2,000 permutations. Operation-stack query-aware AP exceeds the 95% null on 6 of 6 tasks, with median observed-minus-null AP 0.0759; 30% budget recall exceeds the null on 5 of 6 tasks, with median delta 0.0904. These results show that the main AP and budgeted-recall signals are not explained by prevalence and group sizes alone. The same audit is also a boundary: top-five precision exceeds the null on only 3 of 6 tasks,

work-to-first-positive on 0 of 6, width-only operation-stack AP on 5 of 6, and fixed-session/raw-action AP on 6 of 6. R331 therefore supports mechanism isolation and baseline realism, not single-view dominance or a label-free deployment ranker.

R332 asks whether a single visible view/depth policy can be fixed after R320. It reuses the R320 visible policy scores, does not rerun profiling, and excludes oracle policies from selection. Best visible AP is split across operation-stack, fixed-session, and dataset-native views: 3 of 6, 2 of 6, and 1 of 6 tasks, respectively. Best top-five F1 is more dispersed: raw-action stacks win 2 of 6 tasks, while dataset-native, fixed-session, flat, and operation-stack each win 1 of 6. Operation-stack query-aware is best for AP on 3 of 6 tasks and best for 30% budget recall on 3 of 6 tasks, but only 1 of 6 for top-five F1 and 2 of 6 for work-to-first-positive. It still reduces group fragmentation relative to fixed-session query-aware on 4 of 6 tasks, with median group ratio 0.5543. Leave-task source selection does not solve view choice universally: selected-equals-best is 0 of 6 for AP and 0 of 6 for top-five F1. R332 therefore supports the design implication that view, stack fields, predicates, and rankers must remain query-time configuration over the same operations; fixed sessions, span trees, and dataset-native hierarchies remain baselines or drilldown views, not profiler abstractions.

R333 turns the same comparison into an inspection-efficiency curve. It reruns the R320 local scorer over the tracked operation JSONL inputs, yielding 810 visible top-k/work-budget inspection points and 252 task-policy curve rows; it does not fetch, sync, or create data. Within a 30% inspected-work budget, operation-stack query-aware reaches median recall 0.3900, compared with 0.0000 for flat width, 0.3559 for fixed-session query-aware, 0.3377 for dataset-native query-aware, and 0.3325 for raw-action query-aware. At 20% work, the same operation-stack policy reaches median recall 0.2763 versus 0.2422 for fixed-session query-aware. Against flat width, operation-stack query-aware uses lower top-five inspected work on all six tasks and has positive 30%-budget recall where flat has none. Against fixed-session query-aware, it has higher top-five recall on 5 of 6 tasks and fewer groups on 4 of 6 tasks, but lower work-to-first-positive on only 2 of 6 tasks. R333 therefore supports an inspection-cost and fragmentation tradeoff, not dominance on every cost metric.

R334 separates the fragmentation part of that claim from operation-work cost. It reads the tracked R320 policy scores and R333 inspection curves; it does not fetch, sync, create, or relabel data. Against fixed-session query-aware, operation-stack query-aware reduces total groups and positive groups on 4 of 6 tasks, and reaches 50% positive recall with fewer ranked groups on 5 of 6 tasks (median delta -31.5 groups). At the same 30% inspected-work budget, it inspects fewer groups on 5 of 6 tasks (median delta -54.0 groups) while its median recall delta is 0.0275. The same audit preserves the fixed-session work counterpoint: operation stacks have lower work-to-50%-recall on only 1 of 6 tasks, lower top-five work on only 2 of 6 tasks, and lower work-to-first-positive on only 2 of 6 tasks. Thus the supported claim is fixed-session fragmentation reduction and budgeted-recall tradeoff, not work dominance over span-tree drilldown.

Table 3: R320 hidden-label profiler benchmark. Hidden policies read oracle labels and are included only as upper bounds. WTFP is work-to-first-positive.

Policy	Hidden	AP	nDCG	P@5	R@5	F1@5	Work@5	R@30%	WTFP
flat:width	0	0.1678	1.0000	0.1678	1.0000	0.2868	1.0000	0.0000	1.0000
fixed-session:query-aware	0	0.3476	0.7642	0.2555	0.0233	0.0427	0.0163	0.3559	0.0044
dataset-native:query-aware	0	0.3572	0.7445	0.1712	0.8665	0.2822	0.8539	0.3377	0.0582
raw-action:query-aware	0	0.2744	0.5012	0.2513	0.2442	0.2195	0.1507	0.3325	0.0374
operation-stack:width	0	0.1610	0.9364	0.1398	0.4746	0.2147	0.5424	0.3372	0.1694
operation-stack:query-aware	0	0.3117	0.5487	0.1991	0.1880	0.1981	0.0937	0.3900	0.0378
operation-stack:oracle-upper	1	0.5993	0.6372	0.8984	0.3004	0.4029	0.0441	0.5640	0.0122
label-drilldown:oracle-upper	1	1.0000	1.0000	1.0000	0.6703	0.7972	0.1150	1.0000	0.0377

4.3 E3: Mechanism and Actionability

Evidence contract: claim, the profiler exposes actionable knobs in stack fields, mapping/tagging rules, rankers, profile specs, and boundary-derived operation fields; oracle, hidden labels used only after profiling, plus held-out OSWorld-Human boundary positives for R358; baseline, default/patched specs, visible rankers, transfer policies, learned-boundary, fixed-session, and semantic-width stacks; metric, accepted patches, AP delta, top-five lift, first-positive-work, group reduction, and top-five work counterpoint; counterpoint, OSWorld-Human needs boundary fields, so this is not automatic boundary discovery, not automatic patch selection, and not a universal selector.

R320 turns localization scores into an optimization diagnosis rather than only a leaderboard, summarized in Table 4. SATraj unsafe is the cleanest operation-stack win: environment, phase, and action fields plus query-aware risk ranking give the best visible top-five F1. Looping exposes the value and limit of repeat_signal: it helps ranking, but positives are prevalent, so prevalence-aware ranking remains the optimization. Side-effect has large oracle headroom and points to deeper write/input mappings. AgentNet step-quality tasks need action-family localization followed by fixed-session examples. OSWorld-Human needs boundary-derived fields rather than action depth alone. Thus the first actionability result is not that one view wins; it is that the profiler identifies which stack field, mapping, ranker, or drilldown should change for each task family.

The Rust mechanism ablations isolate where those diagnoses come from. Stack-text rules and rank-mode variants (R322/R323) are replayable from profile specs, but they are too coarse for safety and side-effect tasks. Operation-level rank features (R324) close that gap by matching visible mapped operation tokens before folding: semantic-stack feature ranking improves AP over width on 5 of 6 tasks, top-five lift on 4 of 6, and first-positive work on 5 of 6; coarse depth improves AP on 4 of 6 while reducing groups. Leave-one-feature and robustness ablations (R325/R326) make the knobs concrete: repeat_signal, write-action features, and status=success are critical in different families, while misleading safety-like or input-phase features can hurt AP or first-positive work. Equal/global feature banks often preserve the gains, and R325-guided repairs improve AP on 2 of 3 misleading-feature cases, but these repairs use offline scored feedback and are not an automatic learned ranker.

The view and policy audits show the same mechanism at the analysis-surface level. Best AP and best top-five F1 are split across operation-stack, fixed-session, dataset-native, raw-action, and flat views; leave-task source selection cannot reliably pick the best

visible view. Nevertheless, the operation-stack query-aware policy remains Pareto-frontier on all six tasks, beats flat top-five work on all six, beats fixed-session top-five recall on 5 of 6, and reaches fewer groups-to-50% positives than fixed-session on 5 of 6. At a 25% recall target it covers all six tasks with median work 0.2000 versus flat 1.0000 and median 16.0 groups versus fixed-session 50.0; at 50% recall, other depths or views become the best-work counterpoints. Sequence and oracle-depth audits add the inspection unit below and above individual groups: at a 30% operation budget, operation-stack query-aware reaches median 0.4669 positive-session recall versus fixed-session 0.3230, and across 24 oracle-depth rows it has median 0.4342 budget-30 positive-unit recall, beats fixed-session unit recall on 20/24 rows, and reaches 50% positive units with fewer groups on 22/24. The fixed-session depth-gap counterpoint remains, so this is depth-aware triage, not complete intent-boundary recovery. R344 also keeps the metric surface honest: it records 30 supporting baseline-metric comparisons, 16 explicit counterpoints, and nDCG cases where flat or dataset-native views can win.

The actionability portfolio turns those comparisons into reviewer-auditable optimization insight. R335/R341 show that all 36 objective rows have a concrete action, but 27/36 best visible policies are not the default operation-stack query-aware view; transfer misses are mostly view or ranker changes rather than opaque errors. R345/R346/R347 present the same evidence as diagnostic lenses, case packets, and baseline contrasts: six lenses cover 36 objectives, top-five case groups contain positives on 6/6 tasks, top-one groups on 5/6, median top-five lift is 1.6508 at 0.0937 work, and operation stacks beat flat top-five work on 6/6 and fixed-session top-five recall on 5/6 while preserving fixed-session first-positive and flat full-recall counterpoints. R348/R349 then separate visible counterfactual actions from target-label leakage: 36/36 best rows are visible non-oracle, 27/36 require a non-default action, 25/36 require a view change, and held-out action transfer stays within tolerance on 35/60 aligned decisions but exactly transfers the action class only 7 of 60 times and only 2 of 42 times on non-default target actions. R350 packages this into bounded evidence packets: top-five positives on 6/6 tasks, top-one positives on 5/6, strict 30% work budget on 4/6, and the same transfer guardrail. The actionable claim is therefore a tunable diagnosis surface, not a label-free automatic selector.

Finally, executable patches close one loop from diagnosis to profiler configuration. R354 writes default and profile-guided Rust profile specs over the same visible operation input and scores hidden labels only after agentpprof -profile-spec. Five of six patches improve AP, top-five lift, and first-positive work, with median AP delta

0.0376 and median top-five lift delta 0.5750. The rejected OSWorld-Human patch is the important negative case: visible phase/action rank features are insufficient for a human-boundary objective. R358 follows that diagnosis by folding held-out boundary-derived operation fields; learned-boundary profiles improve AP to 0.2583 versus 0.2402 for semantic width and reduce groups to 74 versus 108, while top-five work and first-positive work increase. This supports boundary fields as an actionable operation-field knob, not automatic boundary discovery or automatic patch selection. The result belongs to mechanism/actionability, not a fifth core experiment.

4.4 E4: Reproducibility, Cost, and Claim Integrity

Evidence contract: claim, the offline profiler path and paper claims are reproducible, low-cost, and guarded; oracle, tracked specs/inputs, paper hashes, claim-integrity audits, and reviewer checks; baseline, deterministic versus raw replay, source-status, paper-number, rubric, reviewer, and two-abstraction gates; metric, deterministic pass rate, runtime, number-check and guardrail pass rates, rubric level, and reviewer ACCEPT count; counterpoint, not live eBPF overhead, not human utility, not complete ecosystem compatibility, and not a universal selector.

R338 audits the paper claim itself rather than adding a new empirical result. It reads tracked-clean R320–R350 profiler-evidence artifacts, hashes the current Chinese and English drafts, and checks paper-facing numbers, source policy, must-not-claim guardrails, and the two-abstraction boundary. The audit passes its result-invariant, source-policy, and guardrail checks, so it is a submission-readiness gate for that evidence tranche, not evidence for human utility, automatic boundary discovery, ecosystem compatibility, or a universal selector. R356 then includes R354 and R355 in the full claim-integrity refresh: it checks the R354 5 of 6 patches, 0.0376 AP delta, and 0.5750 top-five lift delta, plus the R355 24 oracle-depth rows, 0.4342 budget-30 unit recall, 20/24 fixed-session recall wins, and 22/24 groups-to-50% wins. This refresh is a paper-consistency gate, not a new empirical result, human-utility claim, or automatic patch selector. R352 then maps the accepted paper state to an OSDI-style profiling-paper rubric. It reads tracked R320–R351 artifacts and current paper text, downloads or syncs no datasets, reruns no profiler, and runs no human or agent analyst task. It passes 26/26 required checks and 10/10 rubric areas, covering fidelity, actionability, baseline tradeoffs, generality, mechanism isolation, robustness/statistics, reproducibility/overhead, and claim-scope guardrails. The result supports the paper's scoped profiler-benchmark readiness; it is not a new empirical result. R360 makes Table 2 reproducible from tracked artifacts. It generates 4 core experiment rows and 20 metric rows, passes 6/6 table checks, and keeps the same non-claims: no profiler rerun, no dataset sync or relabeling, and no human or agent analyst task. Its role is table provenance rather than new evidence. R361 turns the same E1–E4 structure into a claim-evidence ledger. Each core experiment must state the claim, research question, oracle, baselines, primary metrics, headline result, actionable insight, counterpoint, scoped wording, and source runs. The gate passes 10/10 checks, covering hidden-label scale and metric surface, baseline tradeoff rather than

metric dominance, oracle-depth and fragmentation evidence, mechanism/actionability counterpoints, reproducibility and reviewer gates, the two-abstraction boundary, and must-not-claim guardrails. R361 is still a paper-structure artifact: it reruns no profiler, syncs or relabels no dataset, and is not a human or agent analyst task. R363 then packages the E1–E4 evidence into five paper views—baseline tradeoff, metric heatmap, diagnostic lenses, actionability knobs, and oracle-depth adequacy—passing 7/7 checks without making a fifth experiment, rerunning the profiler, or reducing the paper to a flamegraph-only artifact.

Profile-spec determinism and local cost. R327 reruns the existing profile-spec artifacts used by R300, R324, and R326. The workload contains 76 tracked specs: four view specs, 12 operation-feature rank specs, and 60 robustness specs. Each spec is executed twice by the Rust profiler against tracked operation JSONL inputs, for 152 profiler invocations. No dataset is downloaded or synced. The determinism gate hashes the semantic profile content after removing the JSON generated_at timestamp, and separately reports raw-byte hashes.

All 76 specs reproduce the same semantic profile hash, sample count, and unique-stack count across repetitions. Raw-byte determinism is only 4 of 76, because JSON outputs intentionally include a generation timestamp; the stable semantic hash keeps the reproducibility claim on the profile content. Median output size is 37,546 bytes, the largest output is 306,099 bytes, and the largest profile has 2,012 unique stacks. This supports an offline artifact and profile-spec reproducibility claim, not live eBPF overhead or human utility.

R328 closes the raw-artifact caveat with an explicit deterministic-output mode. The Rust CLI accepts `-deterministic-output`, and profile specs may set `deterministic_output`; the mode replaces JSON generated_at and pprof profile time with fixed values while leaving profile content unchanged. R328 reruns the same 76 specs and 152 invocations with this flag. Both semantic and raw-byte determinism are 76 of 76. The clean rerun records empty git and code status. The median runtime in this run is 1,601.0827 ms and the p95 is 2,767.1653 ms; we report these as artifact costs for this run, not as a performance comparison with R327.

5 Related Work and Novelty

Classic profiling and trace analysis. Flame graphs and pprof establish the folded-stack lineage for profiling; profiles consist of weighted samples attached to a location hierarchy, can carry sample labels, and pprof can visualize labels as pseudo stack frames through tag-root or tag-leaf options [1, 2]. Perfetto generalizes trace ingestion and SQL analysis over many trace formats and can derive new events from trace contents [3]. These systems are closer than a simple “fixed hierarchy” baseline. AGENTPPROF therefore does not claim novelty for query-time aggregation alone. The difference is the domain object and evaluation target: the sample is an agent operation, the frames are recursive multi-field operation projections, and the groups are scored against hidden agent-trajectory labels across datasets.

Agent tracing and LLM observability. OpenTelemetry GenAI and OpenInference standardize GenAI spans, LLM calls, agent reasoning steps, tool invocations, retrieval operations, MCP, and

Table 4: R320 actionability summary. Each row identifies the stack or ranker change suggested by the profiler accuracy results.

Task	Best visible policy	Optimization insight
Looping	dataset-native:query-aware	Keep repeat-signal fields but add prevalence-aware ranking because positives are common.
Side-effect	fixed-session:width	Increase write/input weights or use a deeper side-effect mapping before ranking.
Unsafe operation	operation-stack:query-aware	Use environment, phase, and action stack fields and prioritize risky write actions.
Incorrect step	raw-action:width	Use desktop action family first, then drill into fixed sessions for examples.
Redundant step	raw-action:width	Raw action/status currently beats mapped stack depth, so tune quality-specific fields.
Group start	flat:width	Add boundary-derived fields because action-depth stacks fragment starts.

Table 5: R327 profile-spec reproducibility and offline execution cost. Runtime is seconds per spec, measured after the Rust binary exists. Semantic hashes exclude the JSON generated_at field; raw-byte hashes are reported separately.

Workload	Specs	Med. s	P95 s	Med. stacks
R300 view specs	4	3.448	4.273	631.0
R324 rank-feature specs	12	1.539	2.692	41.0
R326 robustness specs	60	1.542	2.640	41.0
All specs	76	1.581	2.719	42.0

Table 6: R327/R328 determinism checks over the same 76 profile specs. R328 uses `-deterministic-output`.

Run	Mode	Sem.	Raw	Med. s	P95 s
R327	default	76/76	4/76	1.581	2.719
R328	deterministic	76/76	76/76	1.601	2.767

provider-specific telemetry [4, 5]. LangSmith, Langfuse, and Phoenix expose traces, sessions, observations, evaluations, dashboards, datasets, and prompt iteration workflows [6–8]. AgentOps gives the closest academic same-problem framing by arguing that agent observability should trace goals, plans, tools, artifacts, and associated data [9]. These systems are strong prior work, so the safe novelty claim is not generic “agent observability.” In this paper, traces, spans, sessions, and observations are exchange containers or baseline shapes. R320 evaluates fixed-session drilldown as the current span-tree proxy; real OpenTelemetry/OpenInference or Phoenix span-tree imports remain future interoperability baselines. The profiler abstractions are only operations and operation stacks, and the central mechanism is that stack shape determines semantic aggregation at query time.

Public labeled agent trajectories. AgentRewardBench, OSWorld-Human, OpenCUA/AgentNet, SATraj-OS, and OSWorld provide the labels and workloads that make R320 possible [12, 18–21]. Prior benchmark papers use these labels to evaluate agents, reward models, safety behavior, or efficiency. We use them as hidden oracles for a profiler benchmark: different stack-shaped aggregates become ranked outputs, and the evaluation measures precision, recall, AP, nDCG, inspection work, fragmentation, and optimization insight. This is the paper’s scoped novelty.

6 Discussion

The result scope is deliberately narrow. The main claim is a profiler claim: on six real labeled tasks, operation-stack profiles can be scored as ranked localization outputs against dataset-provided labels, and they expose a work/fragmentation/actionability trade-off against flat summaries, fixed-session drilldown, dataset-native hierarchy, and raw-action stacks. The supporting audits add uncertainty checks, label-permutation negative controls, view/depth task-fit checks, fixed-recall target costs, sequence-scope scoring, oracle-depth scoring, action counterfactuals, transfer guardrails, and reproducibility gates. They do not add a new empirical claim; they explain when the headline comparison should be trusted and when it should be narrowed.

The strongest negative results define the useful boundary. Operation-stack query-aware ranking improves AP over width on all six R320 tasks, but top-five F1, first-positive work, and high-recall targets remain task dependent. Mapping helps some tasks and hurts others; feature ablations find both critical and misleading fields; held-out action transfer is often within tolerance but rarely transfers the exact action class for non-default targets. Fixed-session drilldown remains a real first-positive counterpoint, raw-action stacks can cover broader session scope, and OSWorld-Human needs boundary-derived fields. These counterpoints are why the design exposes stack fields, mapping rules, rank modes, operation-level rank features, profile specs, and task-specific policies instead of hard-coding one hierarchy.

These results do not show that humans or agents answer questions faster with the tool, nor do they measure live eBPF overhead. They also do not show automatic discovery of all intent boundaries, a label-free automatic selector, or complete compatibility with OpenTelemetry, Phoenix, LangSmith, Langfuse, or Perfetto producers. Those systems remain related work and baselines. In this paper, trace formats are containers; fixed sessions are the evaluated drilldown baseline, while real span-tree imports remain future baselines. The profiler abstractions remain only operations and operation stacks.

7 Conclusion

AGENTPPROF reduces agent profiling to two abstractions. Operations carry all observed and derived fields; operation stacks fold those fields at the depth chosen by the query. Mapping, predicates, rank rules, profile specs, trace exchange, and boundary backends

are mechanisms around those two objects, not additional profiler abstractions.

Across 15 public labeled trace sources, the model covers web, mobile, desktop, software, API, dialogue, failure, safety, human-boundary, and step-quality trajectories. On the four oracle-rich families, operation-stack profiling localizes and ranks labeled problems with less inspection work than flat summaries and less fragmentation than fixed-session drilldown, while the mechanism audits show which fields, mappings, rankers, depths, and profile-spec patches produce the gains. The paper's central claim is therefore a profiler claim: operation/operation-stack profiling exposes dataset-labeled task-relevant failures, quality problems, and semantic boundaries, and produces actionable optimization insight without relying on prompt/session/span boundaries.

The evidence remains scoped. The profile-spec and deterministic-output audits make the offline artifacts reproducible; the claim-integrity and rubric gates keep the paper aligned with tracked results. They do not claim human or agent analyst productivity, live eBPF overhead, automatic discovery of all intent boundaries, a label-free universal selector, or complete compatibility with OpenTelemetry, Phoenix, LangSmith, Langfuse, or Perfetto producers.

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